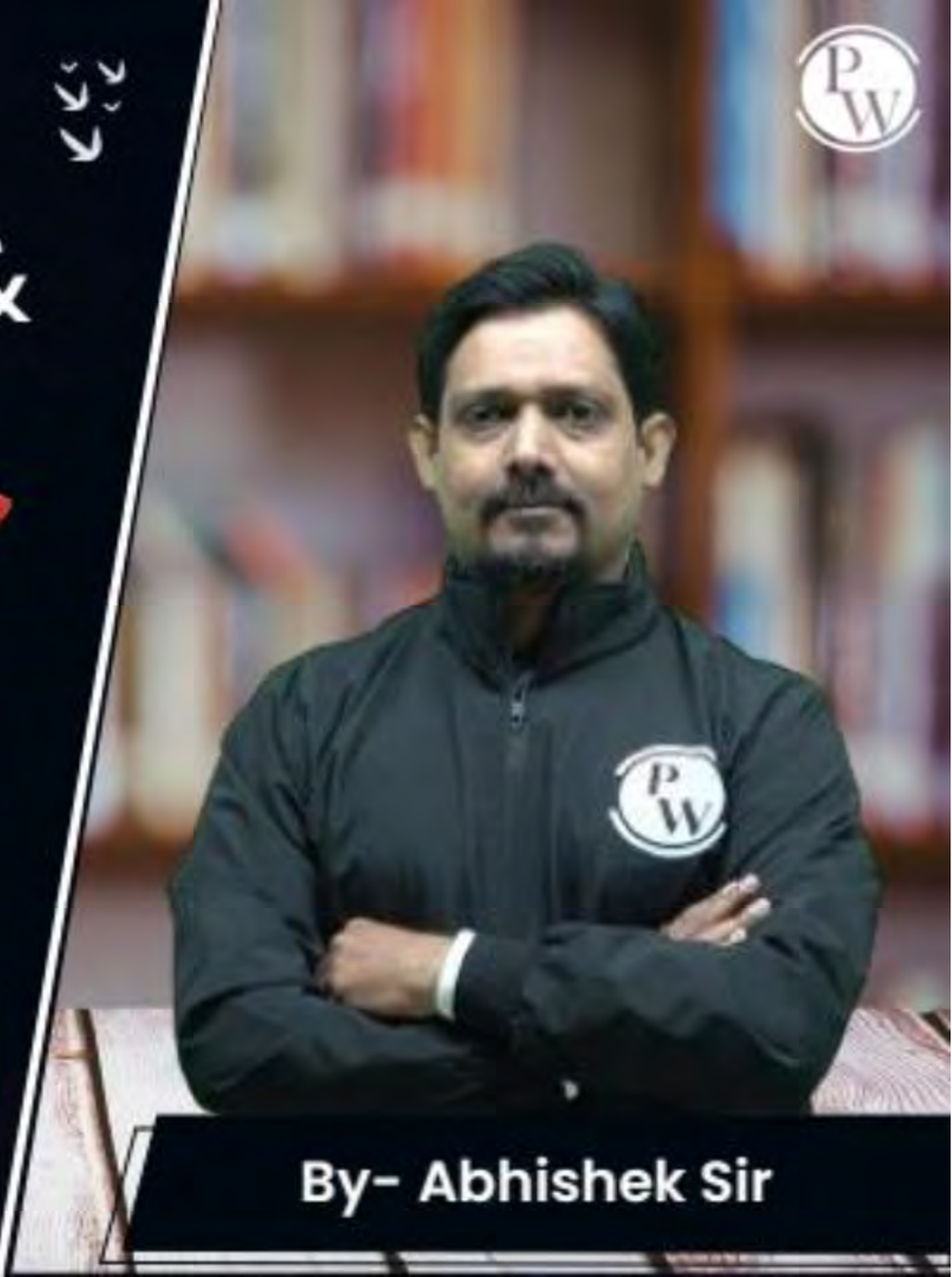
Computer Science & IT



Fundamental of C Language

Introduction

Lecture No. 2



By- Abhishek Sir

Recape of Previous Topic



- Topic
- Topic
- Topic
- Topic
- Topic

System pougoam

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Language - Translation

High level Language

Low level language (Assembly / Machine)

Slide



Topic : System Program

System programs are those programs which Helps in translation of High level Language to low level Language

Slide

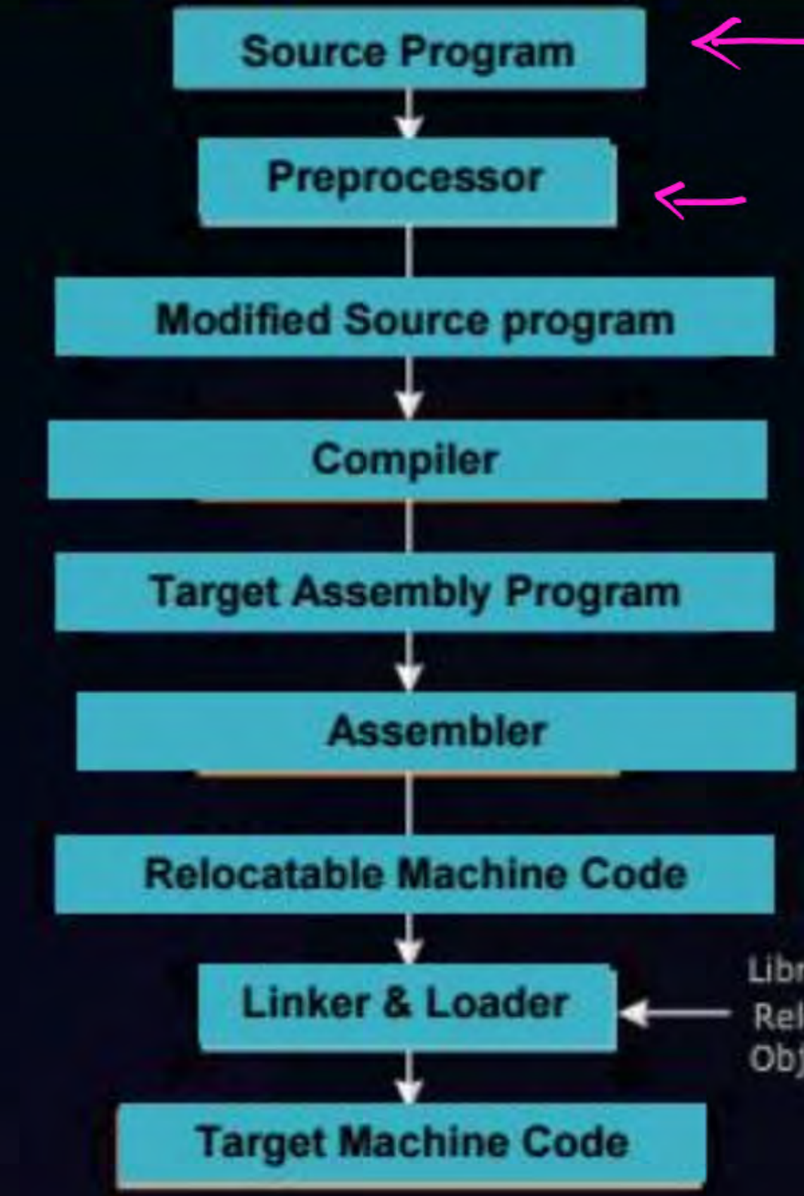


Topic : System Program

The system programs are responsible for the development and execution of a program.



Topic : System Program



Slide



Topic : Preprocessor

Supporting function — Input:
— output:

Input/output file should included in program

macro : `#define Size 100` Syntax

Slide



Topic : Preprocessor

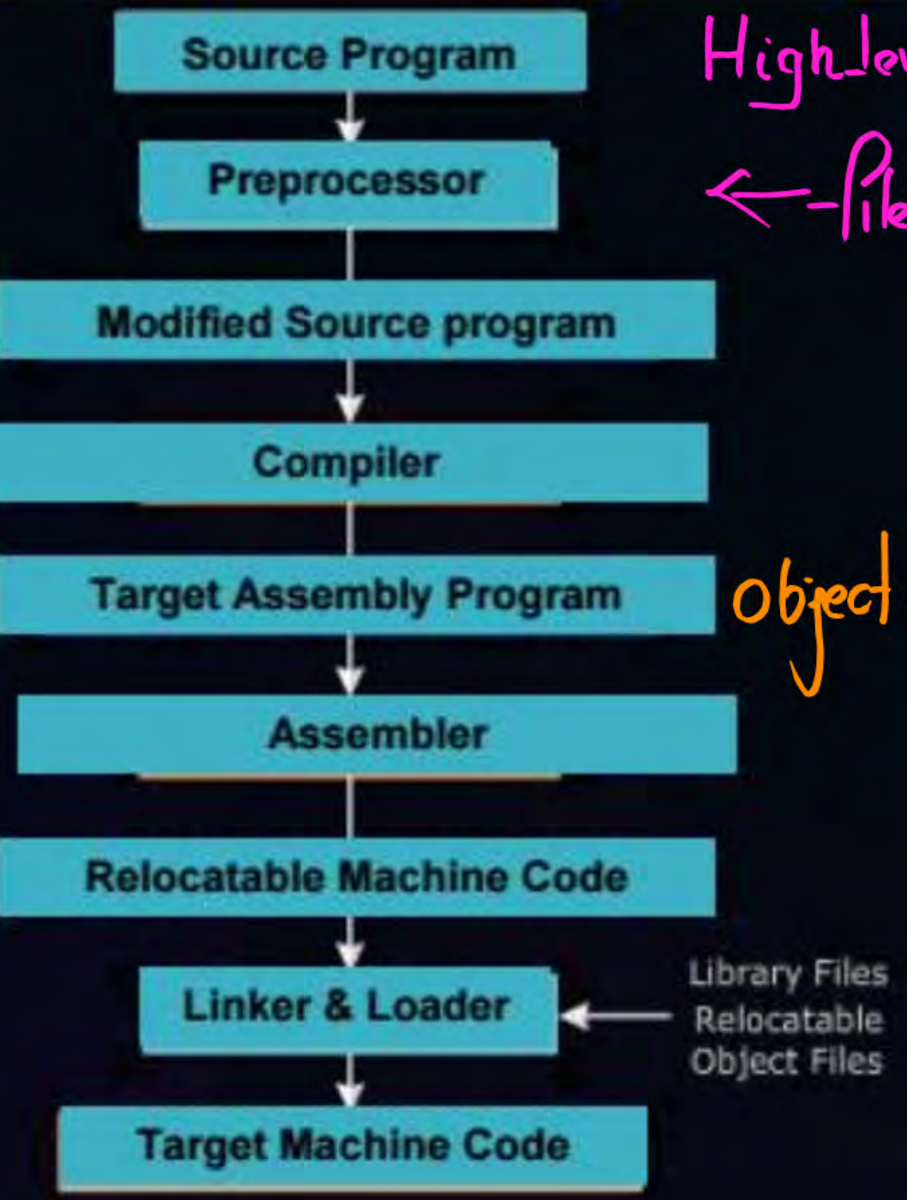
he C preprocessor is a macro processor that is used automatically by the C compiler to transform your program before actual compilation. It is called a macro processor because it allows you to define macros,

macro

Slide

Topic : System Program





High_level

← File Inclusion / Macro expansion

object code

Slide



Topic : Compiler

Modified High level program ⟶ Machine

target Machine

Output of compiler is

Language

object code (Assembly code) Not executable

Syntax Error

Slide



Topic : Compiler

Compilers analyze and convert source code written in languages such as Java, C++, C# or Swift. They're commonly used to generate machine code or bytecode that can be executed by the target host system.



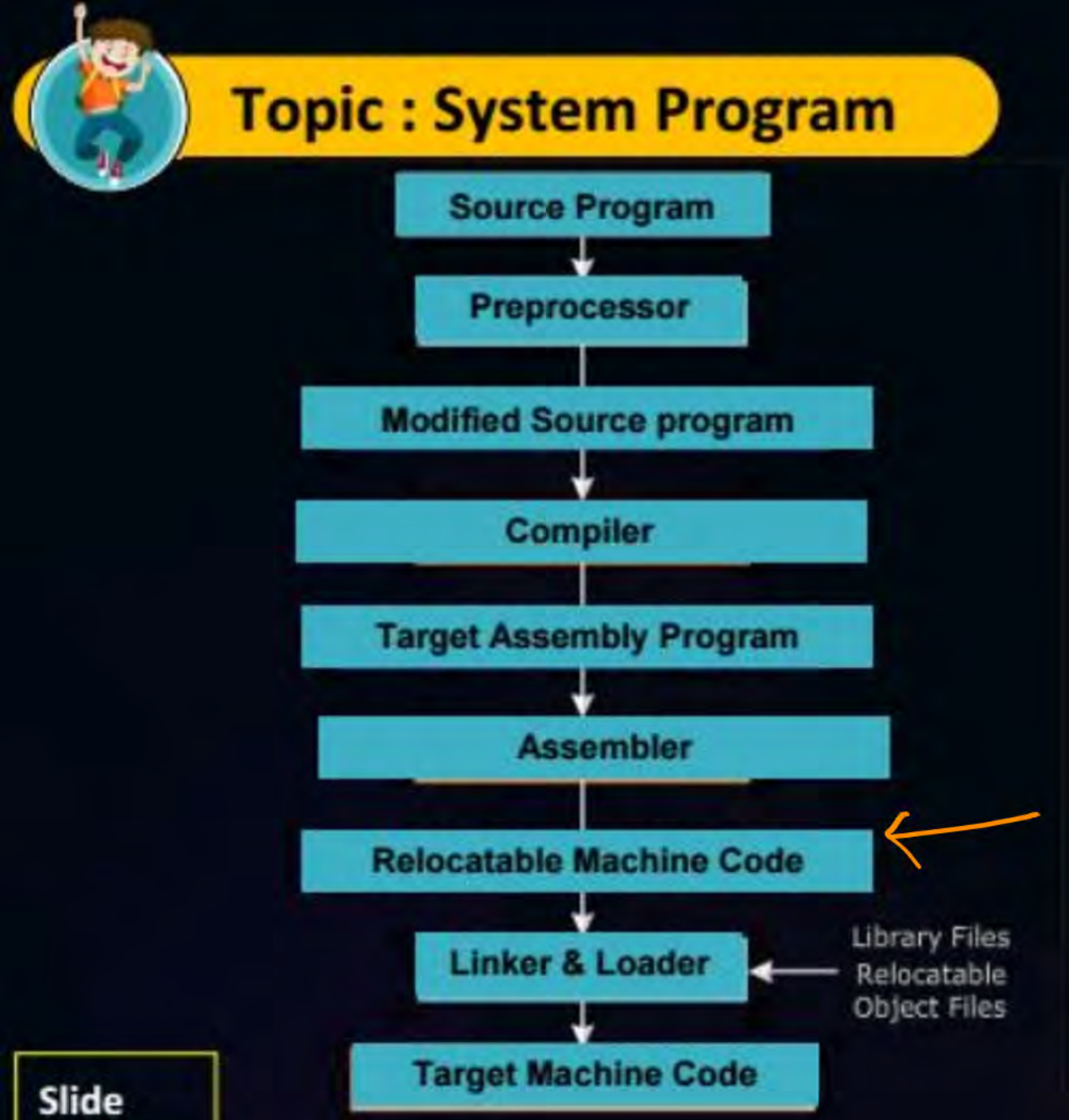
Topic : Assembler

A code (Mnemonic) is given for each instruction on a given m/c. object code uses machine Instruction code used.

Output of Assembler is executable code

Slide

Topic : System Program



Relocatable: It can be executed any part of main memory

Machine

ADD a,b, ⟶ 01101100

00111000

11100111

Slide



Topic : Assembly Language: Low Level Language

We call those languages as low level languages which are closer to hardware as compared to high-level languages instead of software.

they allow programmers to manipulate hardware elements like register, memory etc.

Low-level languages are often used for designing systems, such as developing operating systems, device drivers, and embedded systems.



Topic : Linker

Exe file
Input & output

+

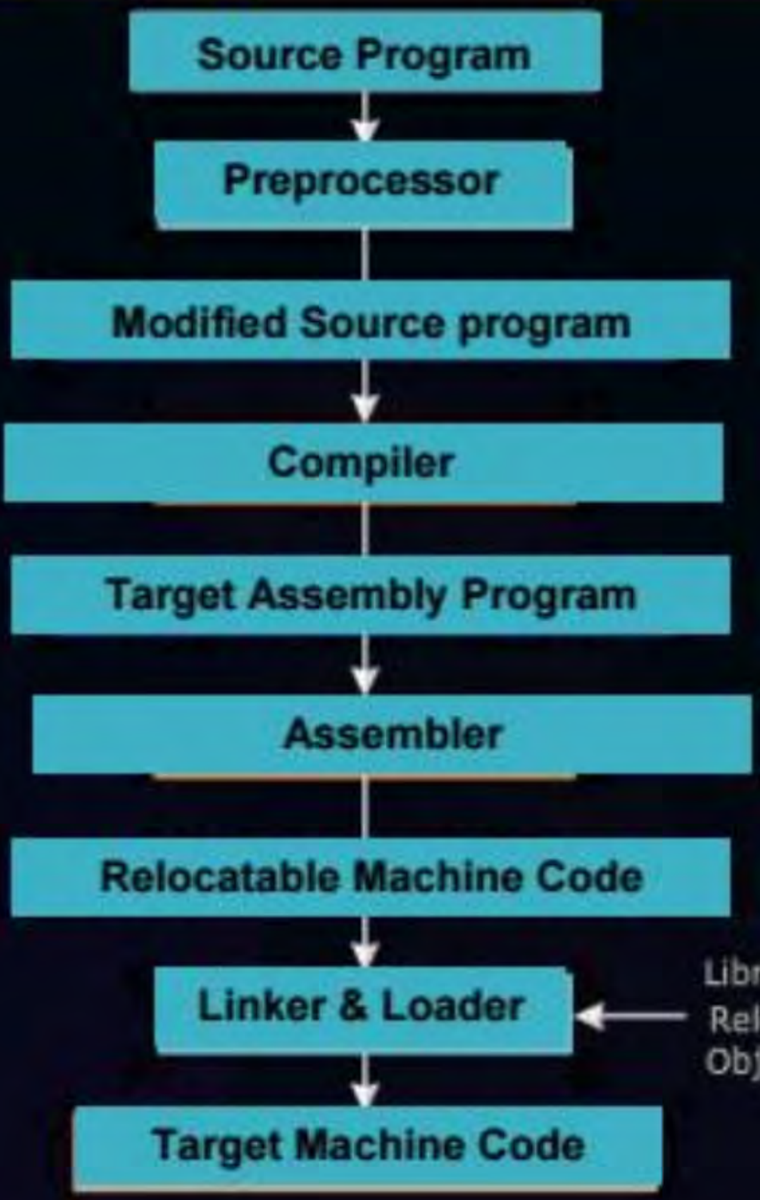
Input/output
Exe file

Combine all files together

Slide



Topic : System Program



Slide



Topic : Linker

A linker is a software tool that plays a crucial role in the compilation process of a program. It takes the object code generated by the compiler and combines it with other necessary libraries and modules to create an executable file.



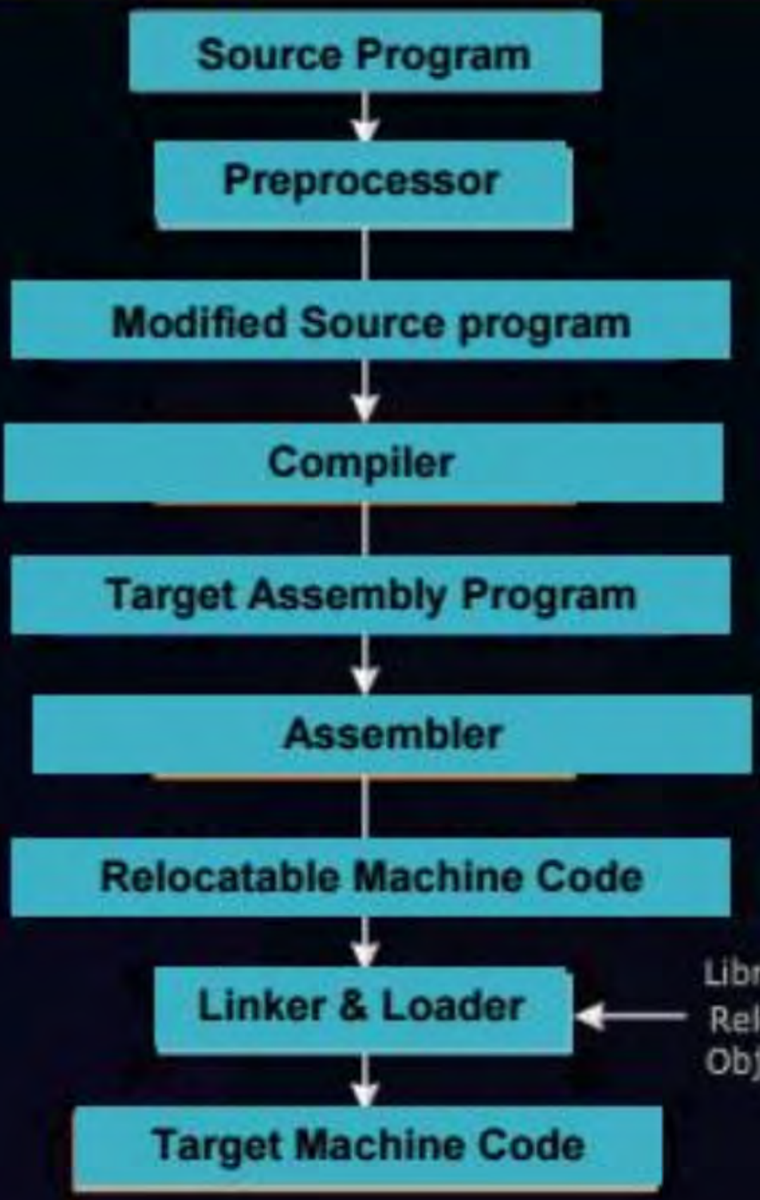
Topic : Loader

Loader is a system program

Loads final executable file in memory for execution



Topic : System Program



Slide



Topic : Loader

The loader loads the program into the main memory for execution of that program. It loads machine instruction and data of related programs and subroutines into the main memory, this process is known as loading.



Topic : Object file

Object file : output of compiler & its consists of target machine language code.



Topic : Executable File

Executable file consists of Machine Instruction code which can be executed by processor

Slide



2 mins Summary

Topic — preprocessor, Compiler

Topic — Assembler, Linker Loader

Topic — object file

Topic — Executable file.

Topic

Slide

THANK - YOU

Slide